

PSEUDOCODE WRITING

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COURSE : CSA0201 - C PROGRAMMING

(1) Electricity Bill Generator

An electricity board charges consumers based on the following:

- * First 100 units - ₹2/unit
- * Next 200 units - ₹4/unit
- * Above 300 units - ₹6/unit

If the total bill exceeds ₹1500, apply a 5% surcharge.

BEGIN

READ name, units

IF units $\leq$ 100 THEN

bill = units * 2

ELSE IF units $\leq$ 300 THEN

bill = (100 * 2) + ((units - 100) * 4)

ELSE

bill = (100 * 2) + (200 * 4) + ((units - 300) * 6)

ENDIF

IF bill > 1500 THEN

surcharge = bill * 0.05

bill = bill + surcharge

ENDIF

PRINT "Consumer Name: ", name

PRINT "Total Payable Amount: ₹ ", bill

END

(2) Smart Attendance System:

A college conducts n classes per day. A student is eligible for the internal exam only if attendance is at least 75%.

BEGIN

READ total-days, attended-days

percentage = (attended-days / total-days) * 100

PRINT "Attendance Percentage: ", percentage, "%"

IF percentage $\geq$ 75 THEN

PRINT "Status: Eligible"

ELSE

PRINT "Status: Not Eligible"

ENDIF

END

(3) ATM Withdrawal Validation:

An ATM allows withdrawal only when $\text{balance} \geq \text{withdrawal}$ amount, withdrawal amount is a multiple of ₹100, minimum balance after withdrawal should be ₹1000.

BEGIN

READ acc-bal, withdrawal

IF withdrawal % 100 $\neq$ 0 THEN

PRINT "Transaction Failed: Amt must be a multiple of ₹100"

ELSE IF withdrawal > acc-bal THEN

PRINT "Transaction Failed: Insufficient balance"

ELSE IF (acc-bal - withdrawal) < 1000 THEN

PRINT "Transaction Failed: Minimum balance of ₹1000 required"

ELSE

acc-bal = acc-bal - withdrawal

PRINT "Transaction Successful!"

PRINT "Remaining Balance: ₹ , acc-balance"

ENDIF

END

(4) Library Fine Calculator:-

```

BEGIN
  READ overdue, mem_sta
  IF overdue <= 5 THEN
    fine = overdue * 2
  ELSE IF overdue <= 10 THEN
    fine = (5 * 2) + ((overdue - 5) * 5)
  ELSE
    fine = (5 * 2) + (5 * 5) + ((overdue - 10) * 10)
  ENDIF
  IF mem_sta == "Premium" THEN
    fine = fine * 0.50
  ENDIF
  PRINT "Total fine: ₹ ", fine
END

```

(5) Employee Performance Evaluation

```

BEGIN
  READ attendance, score, feedback
  a = (attendance >= 90)
  b = (score >= 80)
  c = (feedback >= 8)

  IF a AND b AND c THEN
    PRINT "Performance: Excellent"
  ELSE IF (a AND b) OR (a AND c) OR (b AND c) THEN
    PRINT "Performance: Good"
  ELSE
    PRINT "Performance: Average"
  ENDIF
END

```

(6) Supermarket Billing System:-

BEGIN

READ bill, premium_mem

IF bill > 5000 THEN

discount = bill * 0.10

bill = bill - discount

ENDIF

IF premium_mem == TRUE THEN

add_discount = bill * 0.05

bill = bill - add_discount

ENDIF

gst = bill * 0.18

final_amt = bill + gst

PRINT "Final Payable Amount: ₹ ", final_amt

END

(7) Parking Management System:-

BEGIN

REPEAT

READ choice // "ENTER" or "STOP"

IF choice == "STOP" THEN

BREAK

END IF

READ hours, is_vip, lost_ticket

IF lost_ticket == TRUE THEN

fee = 500

ELSE

IF hours <= 2 THEN

fee = hours * 30

ELSE IF hours <= 5 THEN

fee = (2 * 30) + ((hours - 2) * 50)

ELSE

fee = (2 * 30) + (3 * 50) + ((hours - 5) * 80)

END IF

```

    IF is_vip == TRUE THEN
        fee = fee - (fee * 0.20)
    ENDIF
ENDIF
PRINT "Parking Fee: ₹ ", fee
UNTIL choice == "STOP"
END

```

(8) Online Examination Result Analyzer.

```

BEGIN
    pass_count = 0
    fail_count = 0
    top_score = -1
    topper_name = " "

    FOR student = 1 TO 100 DO
        READ name, mark1, mark2, mark3, mark4, mark5
        total = mark1 + mark2 + mark3 + mark4 + mark5
        average = total / 5
        percentage = average

        IF mark1 >= 40 AND mark2 >= 40 AND mark3 >= 40 AND
            mark4 >= 40 AND mark5 >= 40 THEN
            pass_count = pass_count + 1
            grade = "PASS"
        ELSE
            fail_count = fail_count + 1
            grade = "FAIL"
        ENDIF
    END

```

```
IF percentage > top-score THEN
```

```
    top-score = percentage
```

```
    topper_name = name
```

```
ENDIF
```

```
ENDFOR
```

```
PRINT "Topper Name: ", topper_name, "with", top-score, "%"
```

```
PRINT "Pass percentage: ", (pass-count/100)*100, "%"
```

```
PRINT "Total Passed: ", pass-count, " | Total Failed: ",  
fail-count
```

```
END
```

(9) Smart Water Tank Monitoring System:-

```
BEGIN
```

```
    motor_on = 0
```

```
    motor_status = "OFF"
```

```
    FOR hour = 1 TO 24 DO
```

```
        READ water_level
```

```
        IF water_level < 0 OR water_level > 100 THEN
```

```
            PRINT "Alert: Sensor Error at Hour ", hour
```

```
        ELSE IF water_level < 20 THEN
```

```
            IF motor_status == "OFF" THEN
```

```
                motor_status = "ON"
```

```
                motor_on = motor_on + 1
```

```
            ENDIF
```

```
            PRINT "Hour ", hour, ": Motor turned ON"
```

ELSE IF water_level > 90 THEN

motor_status = "OFF"

PRINT "Hour ", hour, ": Motor turned OF "

ENDIF

ENDFOR

PRINT "Total times motor was switched ON: ", motor_on

END

(10) College Admission Processing System

A college receives applications for admission.

Eligibility:

* Mathematics ≥ 70

* Physics ≥ 60

* Chemistry ≥ 60

* Entrance Score ≥ 80

BEGIN

total = 0

REPEAT

READ continue // "yes" or "NO"

IF continue == "NO" THEN

BREAK

ENDIF

READ name, math, phy, chem, ent, percent

IF math ≥ 70 AND phy ≥ 60 AND chem ≥ 60 AND ent ≥ 80 THEN

total = total + 1

Status = "Admitted"

IF percentage ≥ 90 THEN

scholarship = "100%"

ELSE IF percentage ≥ 80 THEN

scholarship = "50%"

ELSE IF percentage ≥ 70 THEN

scholarship = "25%"

ELSE

scholarship = "0%"

ENDIF

PRINT name, "-", status, "|scholarship: ", scholarship

ELSE

PRINT name, "- Not Eligible"

ENDIF

UNTIL continue == "NO"

PRINT "Total Students Admitted: ", total

END